

3DM Homework Hints and Solutions

Graph Theory Chapter 4

4.2: Use Theorem 16.

4.3: This is not true. It should be relatively easy to find such a graph where $m = n$.

4.5: Discussed in class.

4.6: Use Theorem 18 and Theorem 1.

4.7: Discussed in class.

4.8: Discussed in class.

4.9: Usually, one would prove a collection of statements like this equivalent by proving that $a \implies b \implies c \implies d \implies e \implies a$. But in this case, we already know from Theorems 18 and 21 that $a \iff c$ and $a \iff d$, i.e., we already know that a , c , and d are equivalent statements. So all we have to do is prove that b and e are each equivalent to one of a , c , and d . I suggest you do this by proving that $a \iff b$ and $d \iff e$.

4.10: I suggest you prove $a \implies b \implies c \implies d \implies a$.

4.12: There are simple counterexamples to both statements.

4.13: ($\implies$) Suppose that G is a forest. Let G' be an induced subgraph of G (note that G' is also a forest) and C' a component of G' (hence, C' is a tree). If C' is trivial, then the single vertex in C' has degree 0; on the other hand, if C' is nontrivial, then by Theorem 16, C' has a vertex of degree 1.
($\impliedby$) We'll prove the contrapositive: Suppose that G is not a forest. Then G contains a cycle, C . But then the subgraph of G induced by $V(C)$ has minimum degree 2 or more, completing the proof.

4.14: Discussed in class.